Assignment-1

Java Compilation and Bytecode Analysis

Objective

The objective of this assignment is to understand the **Java compilation process**, execution of Java programs, and **bytecode disassembly using** `javap`. Students will write simple Java programs, compile them, execute them, and inspect the generated bytecode.

Part 1: Write and Execute Java Programs

Write the following Java programs.

1. Add Two Numbers

Write a program that reads two integers from the user and prints their sum.

2. Fibonacci Series

Write a program to print the **first 10 numbers of the Fibonacci series**.

Example:

```
0 1 1 2 3 5 8 13 21 34
```

3. Print Your Name 10 Times

Write a program that prints **your name 10 times using a loop**.

4. Factorial of a Number

Write a program that calculates the **factorial of a given number**.

Example:

```
Enter number: 5
Factorial = 120
```

5. Check Even or Odd

Write a program that checks whether a given number is **even or odd**.

6. Largest of Three Numbers

Write a program to find the **largest among three numbers**.

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7. Reverse a Number

Write a program that **reverses the digits of an integer**.

Example:

```
Input: 1234
Output: 4321
```

8. Multiplication Table

Write a program that prints the **multiplication table of a given number** up to 10.

Example:

```
5 x 1 = 5
5 x 2 = 10
...
```

9. Sum of First N Natural Numbers

Write a program to compute the **sum of the first N natural numbers**.

Example:

```
Input: 10
Output: 55
```

10. Swap Two Numbers

Write a program to **swap two numbers without using a third variable**.

Part 2: Compilation and Execution

For each program perform the following steps in the terminal.

Compile the program

```
javac ProgramName.java
```

Run the program

```
java ProgramName
```

Take **screenshots of the console output**.

Part 3: Bytecode Analysis Using javap

After compiling the program, analyze the `.class` file using:

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Basic information

```
javap ProgramName
```

Display bytecode instructions

```
javap -c ProgramName
```

Verbose output

```
javap -v ProgramName
```

Observe the following:

- JVM instructions (`iload`, `istore`, `iadd`, etc.)
- Method signatures
- Constant pool entries
- Stack and local variable information

Part 4: Report Preparation

Prepare a **PDF document** containing:

- Source code of all programs
- Compilation command (`javac`)
- Execution command (`java`)
- Console output screenshots

  Output of:

  ```
  javap

  javap -c

  javap -v
  ```

All screenshots must be **from the actual terminal/command prompt**.

Submission Guidelines

Format: **PDF**

File naming format:

```
Name_JavaBytecodeAssignment.pdf
```

Example:

```
RoodranshuNagotiya_JavaBytecodeAssignment.pdf
```

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